

Geocomputation & Machine Learning course

Quick recap

This was a GRASS GIS training session where Professor Giuseppe taught students the fundamentals of using GRASS for geographic information system operations. The class covered GRASS software architecture, command-line syntax, and practical exercises including creating locations, map sets, and performing raster operations like slope calculations. Students learned about the GRASS database structure, working with temporary folders, and using the graphical user interface versus command-line approaches. The session included troubleshooting tips for data downloading and handling, with students asking questions about using PKTools for raster correlation, wget commands for MODIS data, and working with different data types and resolutions. Professor Giuseppe emphasized the importance of properly preparing raster datasets with consistent extent, resolution, and projection before using GRASS to avoid alignment issues. The class concluded with information about an upcoming hackathon week in Matera where students will work on their own projects and present their code and modeling work.

Next steps

Annabell

- Write an email to Giuseppe regarding the late check-in at Rock Hostel so he can put her in contact with the hostel for a late-entry code.

Areeba

- Try resampling rasters to a coarser resolution (e.g., 1 km) or using random sampling of cells to create pixel-to-pixel scatter plots in Python or R.

Giuseppe

- Forward Annabell's email to Rock Hostel to arrange a late check-in code.

Collaboration

- Sun,: Test data download commands directly in the Bash terminal instead of Jupyter Notebook to avoid freezing issues.
- All students: Run the provided installation commands in the Ubuntu terminal (git pull and Python package installation) to update the course materials and enable the "system globe" Python environment in JupyterLab.
- All students: Practice creating a GRASS location and map set from a reference raster file, performing a raster operation (e.g., slope or aspect), and exporting the result using `r.outgdal`.
- All students: Prepare their own datasets (ensure correct projection, pixel size, and no-data values) and try to implement a GRASS-based processing script before the next session.
- All students: Attend the next session on Thursday for hydrological modeling and advanced GRASS scripting.
- All students: Attend the catch-up session next week to discuss their specific data analysis problems, scripts, and progress.
- All students: Complete the geocomputation part of their project before arriving in Matera for the Hackathon week.

Summary

Data Integration and Processing Challenges

Giuseppe and Andrea discussed data integration challenges, with Andrea explaining that price and fuel registration data from 2015 to the third quarter of 2026 are stored in separate files that need to be combined. Areeba asked about using PKTools for raster correlation and pixel-to-pixel scatter plots, and Giuseppe suggested using Python with resampling techniques to handle large datasets more efficiently. Sun reported issues with downloading data using WGET in Jupyter Notebook but found better success with CURL, and Giuseppe recommended testing commands directly in the terminal rather than Jupyter Notebook environment. Giuseppe also demonstrated how to use GDAL commands to download specific bands from MODIS data and crop files to specific regions rather than downloading the full globe.

GRASS GIS Introduction Overview

Giuseppe provided an introduction to GRASS, explaining its history as a GIS software starting in the 1970s, its integration with other tools like QGIS and R, and its capabilities for raster and vector operations, hydrological analysis, and 3D image processing. He detailed GRASS's folder structure, including the database, location/project, and map set components, emphasizing the importance of properly preparing datasets before using GRASS. The discussion covered GRASS's command syntax, which follows Bash syntax, with different prefixes (G, R, V) for general, raster, and vector operations respectively.

GRASS GIS Demonstration Session

Giuseppe demonstrated how to use GRASS GIS, focusing on both the graphical user interface and command-line options. He explained the three-step process for working with GRASS and showed how to navigate the interface, including the new single-window design. Giuseppe discussed the importance of using the correct version of GRASS and its manual, and he emphasized the utility of using the command line for scripting, particularly for HPC environments. The session included a brief overview of the GRASS command structure and how to access help documentation within the system.

GRASS GIS Directory and Map Sets

Giuseppe explained the concepts of working directories and map sets in GRASS GIS, demonstrating how to use commands like ``print working directory`` and ``g.list`` to navigate and work with different map sets. He emphasized the importance of setting the G region for defining the study area and controlling computation speed, particularly for testing scripts and optimizing processing time. Giuseppe also showed how to change the pixel resolution and explained its practical applications for both spatial analysis and performance optimization.

GRASS GIS Implementation Discussion

Giuseppe explained the importance of using the full path when starting GRASS, emphasizing that users should always provide the full path to avoid starting in an unknown location. He highlighted GRASS's capabilities in hydrological modeling, terrain analysis, and satellite image segmentation, mentioning ongoing developments in integrating machine learning into the

platform. The group took a break and planned to continue the discussion after 30 minutes, with Giuseppe preparing to demonstrate using GRASS from a virtual machine.

Hydrological Modeling with GRASS GIS

Giuseppe demonstrated a hydrological modeling technique using GRASS GIS software, specifically showing how to snap points to a river stream network using R. He guided the participants through creating a GRASS project from scratch using a digital elevation model of South America, explaining the importance of properly structured raster environments with defined pixel resolution and no-data values. The demonstration included creating a temporary location and map set, importing a TIFF file, calculating slope from the elevation data, and exporting the results using GDAL. When asked if slope information would be lost when saving as a 16-bit format, Giuseppe confirmed that some precision would be lost but the information would still be preserved.

GDAL and GRASS Data Handling

Giuseppe explained the importance of understanding data types when working with GDAL and GRASS, particularly when dealing with float numbers versus integers, noting that using float types can result in significantly larger file sizes. He demonstrated the difference between temporary folders that are automatically deleted after computation and permanent locations that can be accessed later, showing how to create and use permanent locations with proper projection information. The discussion included practical examples of creating slope and aspect calculations that can be saved and accessed through both command line and GUI interfaces.

GRASS GIS Command Line Demonstration

Giuseppe demonstrated how to use GRASS GIS commands through the command line interface, showing how to perform resampling operations and work with digital elevation models for hydrological modeling. He explained the difference between using the TMP directory (which stores temporary files in RAM) versus regular disk storage, noting that using the TMP directory can significantly speed up computations. Abdul confirmed he was not familiar with this command-line approach but would try it, while Shiva mentioned previous challenges with the graphical interface losing temporary files. Francesco asked about downloading data and was directed to use the git clone command for access to the necessary files.

GRASS Memory Management Training

Giuseppe explained how to manage RAM usage when working with large files in GRASS, emphasizing the importance of monitoring memory usage to avoid performance issues. He demonstrated how to install and use GRASS add-ons for various operations including area calculations and parallel processing. The group discussed upcoming activities, with Giuseppe announcing that next week will be a catch-up session where students can share their progress and challenges with geocomputation work, while the Matera hackathon week will focus on students working with their own data with instructor support.